From Multi-Agent Simulation to Causal Multilevel Graphs in Parkinson's Disease

DCC-MASL 25–26

Alessandro Massimo Mermali (136584)
alessandroma.mermali@studenti.unicam.it

University of Camerino, Italy

Presentation Goal

Guiding question

How can selected mechanisms of Parkinson's disease progression be translated into a distributed simulation whose execution can be observed and interpreted at multiple causal levels?

Biology

mechanisms → feedbacks

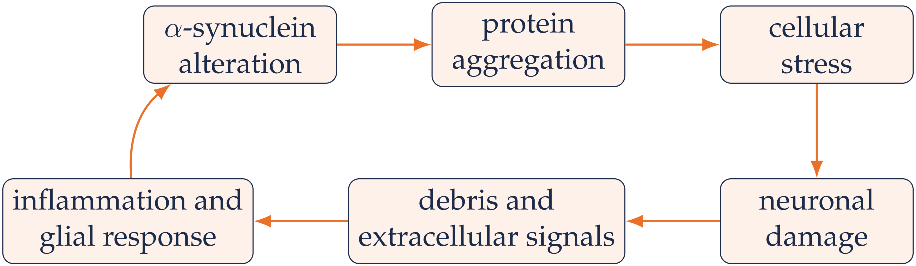
Model

agents → local rules

Analysis

causal trace → abstraction levels

Modeled Pathological Loop



Modeling scope

The project does not aim to reproduce Parkinson's disease as a whole. It makes a selected set of interacting pathological mechanisms explicit and executable.

Biological Mechanisms Represented in the Model

Intracellular mechanisms

- α-synuclein misfolding and aggregation, formation of Lewy bodies
- Mitochondrial stress, damage and recovery
- Lysosomal targeting and degradation
- Accumulation of intracellular damage
- Transitions between neuronal health states

Extracellular mechanisms

- Dopamine release by healthy neurons
- Transmission of extracellular α-synuclein
- Accumulation and clearance of extracellular debris
- Microglial activation
- Astrocytic support and reactive response

From Biology to an Agent-Based Model

Biological concept	**Model representation**
Dopaminergic neuron	macro-agent with internal state
α-synuclein	protein agent with state transitions
Aggregate / Lewy body	entity collecting protein members
Mitochondrion and lysosome	intracellular agents
Microglia and astrocyte	extracellular reactive agents
Substantia Nigra	spatial environment and global fields
Biological mechanism	local rule, transition or interaction

Methodological idea

Each agent observes a limited context, updates its state and acts locally. System-level progression is not directly scripted: it emerges from interactions among agents and fields.

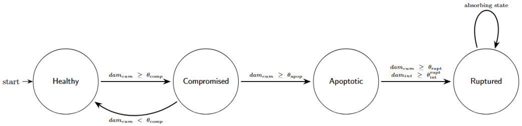
Adaptive Agents

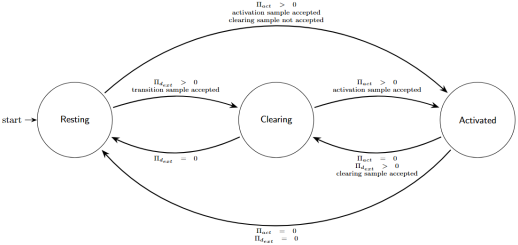
- Agents are modeled as **adaptive agents**.
- Their actions depend on their internal state and perception.
- No global controller declared, the behavior logic emerges from the local entities.
- The environment works only as a bridge between the agents.

An adaptive agent is defined as:

- Sets:
  - I, internal states
  - E, environmental states
  - Per, perceptions
  - Ac, possible actions
- i_0, initial state
- Functions:
  - $see : E \to Per$
  - $next : I \times Per \to I$
  - $action : I \times Per \to Ac$
  - $do : Ac \times E \to E$

Some agents example





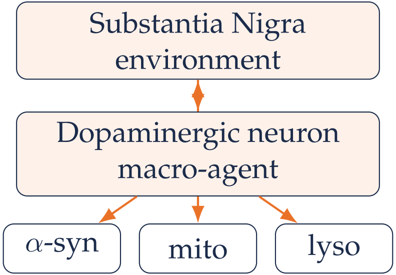
Nested Multi-Agent Structure

Extracellular level

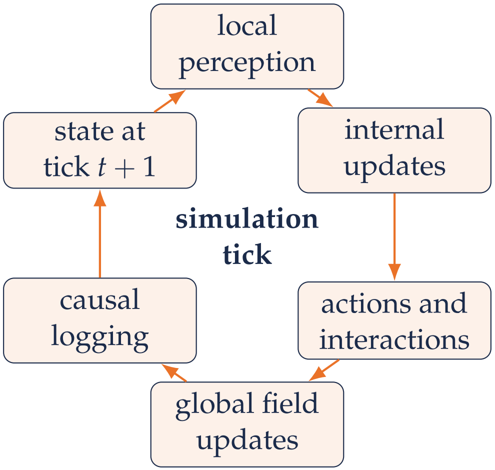
The Substantia Nigra contains dopaminergic neurons, glial cells, extracellular α-synuclein, debris, inflammation and dopamine fields.

Intracellular level

Each neuron acts as a macro-agent: it contains proteins, mitochondria, lysosomes and internal scalar fields.



SIMULATION TICK



Observability and Causal Logging

What is recorded

- agent and environment states over time;
- transitions and interactions;
- causal relations between states, actions and effects.

Why it matters

The simulation does not only produce trajectories. It also produces an explanatory trace that can be transformed into graphs for post-run analysis.

$\text{Neuron12\_Healthy@40} \xrightarrow{\text{transition}} \text{Neuron12\_Compromised@41}$

How to Read a Simulation Run

Pathology progression

- healthy neuron loss;
- compromised and apoptotic states;
- rupture events.

Intracellular balance

- free vs aggregated α-synuclein;
- lysosomal clearance;
- mitochondrial damage.

Environmental feedback

- debris accumulation;
- inflammation growth;
- dopamine reduction.

Post-Run Multilevel Analysis

G_0 causal trace → G_1 time-contracted → G_2 agent-clustered → G_3 pattern graph

Main idea

Start from the full timed causal trace and progressively remove details that are not needed at the next scale, while preserving interpretable causal information.

Graph Levels and Interpretation

Level	Contraction principle	Interpretation
G_0	none	full execution trace
G_1	remove time index	recurring state interactions
G_2	merge agent identities	agent-type mechanisms
G_3	contract structural patterns	relations among causal motifs

G_3 contribution

G_3 abstracts from individual agent states to higher-order structures such as cycles, strongly connected components, repeated causal paths and dense interaction motifs. Its goal is to expose the recurring organization of disease-driving mechanisms rather than single events.

G_0 DEFINITION: TEMPORAL CAUSAL GRAPH

- **Node definition:** either an agent in a state at a time (`AgentNameID_State@time`) or an environment field at a time (`EnvironmentField@time`)
- **Edge definition:** directed edges encode causal influence.
  - agent at moment $t \rightarrow$ agent at moment $t + 1$;
  - agent $\rightarrow$ environment variable;
  - environment variable $\rightarrow$ agent;

Simplification

For simplification purposes, some mechanisms were directly represented at agent level.

- some agent transition;
- lysosome targeting relation;
- aggregation;

NOTE: G_0 is always acyclic.

G_1: Time-Contracted Graph

Contraction rule

Collapse nodes that differ only by time: `x@1`, `x@2`, `x@3` $\mapsto$ `x`

Analytical role

G_1 captures recurring causal relations between states of the same concrete agents and environment fields, independent of when they occur. It can identify state-level feedback loops and recurrent causal dependencies.

Consequence

Removing time can create cycles, because repeated temporal interactions are projected onto the same node set.

G_2: Agent-Clustered Graph

Contraction rule

Collapse individual agent identities while preserving agent type (`A`) and state (`S`):

$$\texttt{A1\_S},\ \texttt{A2\_S} \quad \mapsto \quad \texttt{A\_S}$$

Analytical role

G_2 reveals how classes of agents interact with each other and with environmental variables, independently of specific agent instances.

- Removes individual IDs.
- Aggregates repeated mechanisms across many agents.

G_3: Structural Pattern Graph

Contraction rule

Contract relevant topological structures in G_2, such as:

- cycles;
- cliques;
- strongly connected components;
- repeated causal paths;

Interpretation

G_3 is a graph of patterns. Nodes no longer represent individual states, but higher-order causal structures.

What the Abstraction Preserves

Preserved information

- source and target mechanism;
- interaction type;
- frequency of repeated relations;
- first and last tick of occurrence;
- aggregated effect sign and intensity.

Interpretation

The abstraction does not simply simplify the visualization. It turns an execution history into a hierarchy of explanations: from event-level causality to system-level motifs.

Monomer@t → Misfolded@t+1 → Oligomer@t+k

Conclusion

What the project offers

- explicit and inspectable mechanisms;
- connection between local events and global dynamics;
- support for parameter experiments;
- post-run analysis across causal scales.

What it does not claim

- clinical prediction;
- definitive biological calibration;
- complete disease coverage;
- quantitative validation on experimental data.

The main result is not only a disease progression simulator, but an infrastructure to ask how that progression emerges.